

Total No. of Questions : 8]

SEAT No. :

P-6671

[Total No. of Pages : 2

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B.E. (Mechanical Engineering)

PRODUCT DESIGN AND DEVELOPMENT

(2019 Pattern) (Semester - VII) (402045A) (Elective - IV)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4 Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data if necessary.

- Q1)** a) Explain SWOT analysis in selecting profitable product concept. [7]
b) Describe in detail reverse engineering. [6]
c) Explain Economic analysis in product analysis. [4]

OR

- Q2)** a) Explain concept analysis considering functional, marketing, operational etc. aspect. [7]
b) Which are the different idea generation approaches use in product design and development. [6]
c) What is product policy of an organization? List down various product policies. [4]

- Q3)** a) What is Ergonomics in design? Explain types of Ergonomics with example. [7]
b) Explain BOM with example. [6]
c) Define Limit, Tolerance and Fit. [4]

OR

- Q4)** a) What is product architecture? Explain types of product architecture. [7]
b) What is the need for engineering drawing? Classify engineering drawing. [6]
c) What is part print analysis? How it is important in product development. [4]

P.T.O.

- Q5)** a) What is production capacity planning? Explain the steps followed in planning. [8]
b) What is Rapid prototyping? Define and enlist various methods of prototyping. [6]
c) Define letter of intent, purchase order and product costing in vendor development. [4]

OR

- Q6)** a) Explain stereolithography in detail with suitable sketch. [8]
b) Explain FEA with example. [6]
c) Why homologation certificate is important in design and development? Explain with example. [4]

- Q7)** a) What is APQP, what role it plays in robust design and development? [8]
b) Compare Value analysis and Value engineering. [6]
c) Discuss the elements of PLM in detail. [4]

OR

- Q8)** a) Explain application of Teamcenter/Nx Manager in Product design and development. [8]
b) Compare PFMEA AND DFMEA. [6]
c) What are guidelines for design for robustness? Discuss. [4]
